Bellabeat Case Study Final Report

1. Business Task Statement (Ask Phase)

Bellabeat, a high-tech company that manufactures health and wellness smart products for women, aims to grow its market share and increase engagement. The core business task is to analyze smart device usage data to uncover actionable insights into how consumers are currently using their fitness trackers and apply these findings to influence Bellabeat's marketing strategy for the Bellabeat products.

Analyze smart device usage data from non-Bellabeat devices to uncover behavioral trends, then apply those insights to one Bellabeat product and deliver actionable marketing strategy recommendations to the executive team.

2. Data Sources Description (Prepare Phase)

Dataset Name and Location

The primary dataset used for this analysis is the **FitBit Fitness Tracker Data** (publicly available on Kaggle).I used the data from the Kaggle website without downloading , for this I used OS and json libraries. The data is stored on the kaggle website [Link].

Organization

The data is organized into 18 CSV files, covering various aspects of fitness tracking (daily activity, sleep, weight, heart rate, etc.) for 33 unique users over a 31-day period . Key tables analyzed include `dailyActivity_merged.csv` and `sleepDay_merged.csv`.

ROCCC Evaluation

Attribute	Evaluation	Description
Reliable	Medium	Data is third-party data from an unknown source/collection method; reliability is questionable.
Original	Low	Data is from a third party (FitBit), not Bellabeat's original data.
Comprehensive	Medium	Includes various metrics (steps, calories, sleep, intensity) but only covers a limited time frame (31 days) and a very small user base (33 users).
Current	Low	The data is not current (collected in 2016), which may not reflect current user behavior or product features.
Cited	Yes	The data is cited as public domain from Kaggle and original owners are clearly stated.

Limitations of the Data

The most significant limitations are the small sample size (33 users) and the short observation period (one month), which severely limits the ability to generalize findings to Bellabeat's target audience of women. Furthermore, the data does not contain demographic information, making it impossible to segment users based on age, gender, or location.

3. Data Cleaning Documentation (Process Phase)

Tools Used

The data cleaning and processing were primarily performed using **Python** within the **Google Colab** environment, leveraging the `pandas,numpy,matplotlib, seaborn,os and json` libraries for data manipulation.

What Was Checked

Check	Action Taken
Consistency of Date/Time Format	Standardized date formats using `pd.to_datetime`.
Unique User IDs	Confirmed the number of unique users in each dataset (found 33).
Missing/Null Values	Checked for missing values across all columns. None found in key activity/sleep tables.
Duplicate Rows	Checked and removed duplicate entries to ensure data integrity.

What Was Cleaned

1. **Extra Spaces :** Clean all the extra spaces in the entire data set.
2. **Duplicate Entries:** Removed 3 duplicate rows from the `sleepDay_merged` table.
3. **Date Conversion:** Converted date columns from strings to datetime objects for accurate merging and analysis.

How the Two Tables Were Merged

The `dailyActivity_merged` and `sleepDay_merged` tables were merged using an inner join based on two common keys: `Id` and the date column (renamed to `ActivityDate`). This ensured that only records where a user had both activity and sleep data for a given day were included in the combined analysis dataset, stored as `df_merged`.

4. Analysis Summary (Analyze Phase)

Key Statistics Found

- **Average Steps:** 7,637 steps/day. This is below the widely recommended 10,000 steps.
- **Average Distance:** 5.49 km/day.
- **Average Calories Burned:** 2,303 calories/day.
- **Average Sleep:** 6.98 hours/night.

User Classification Results

Based on the average daily step count, users were classified into four groups:

User Type	Daily Step Range	Percentage of Users
Sedentary	0 - 5,000 steps	24.2%
Lightly Active	5,001 - 7,499 steps	27.3%
Fairly Active	7,500 - 9,999 steps	27.3%
Very Active	10,000+ steps	21.2%

Insight: Almost one-fourth (24.2%) of users are **Sedentary**, representing a significant opportunity for Bellabeat to encourage increased activity.

Trends and Relationships Discovered

- **High Sedentary Behavior —** 51.5% of smart device users fall into Sedentary or Lightly Active categories, with an average of 991 minutes per day spent inactive despite owning a smart device
- **Low Active Minutes —** Users average only 21 very active minutes per day.
- **Steps Drive Calories —** Clear positive relationship between daily steps and calories burned.
- **Weekend Inactivity —** Sunday is consistently the least active day (6,933 avg steps).
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- **Activity vs Sleep Tradeoff —** More active users tend to get slightly less sleep.

Summary Table of Findings

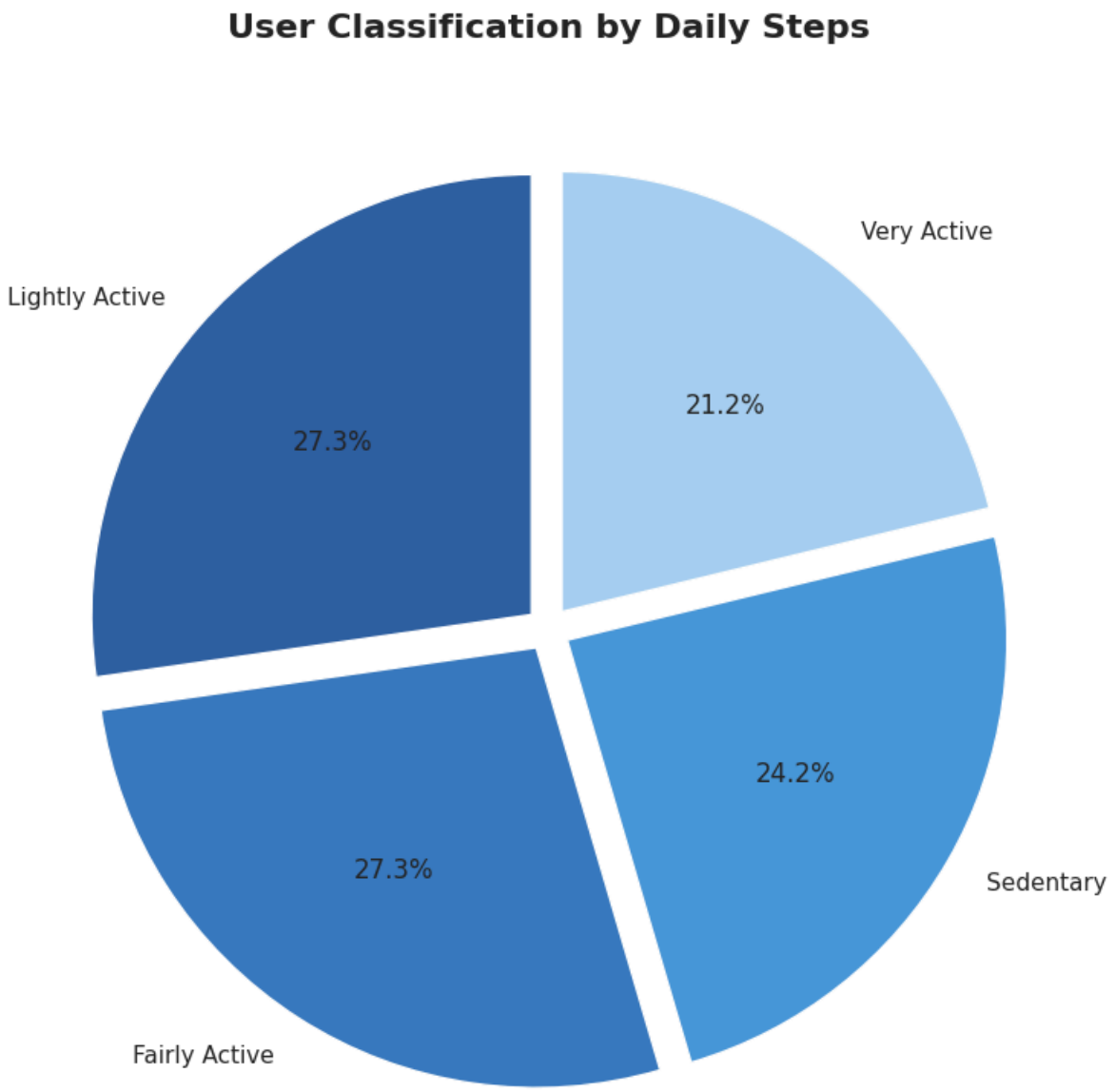
Finding	Impact/Insight
51.5% of users are Sedentary.	Major opportunity for engagement/motivation campaigns.
Activity is lowest on Weekends.	Sunday is consistently the least active day (6,933 avg steps).

Finding	Impact/Insight
Low Active Minutes	Goal setting features and encouraging 10k steps should be emphasized.
Avg. Time Asleep (6.98 hours) is healthy.	Focus on *quality* of sleep and integration with activity.

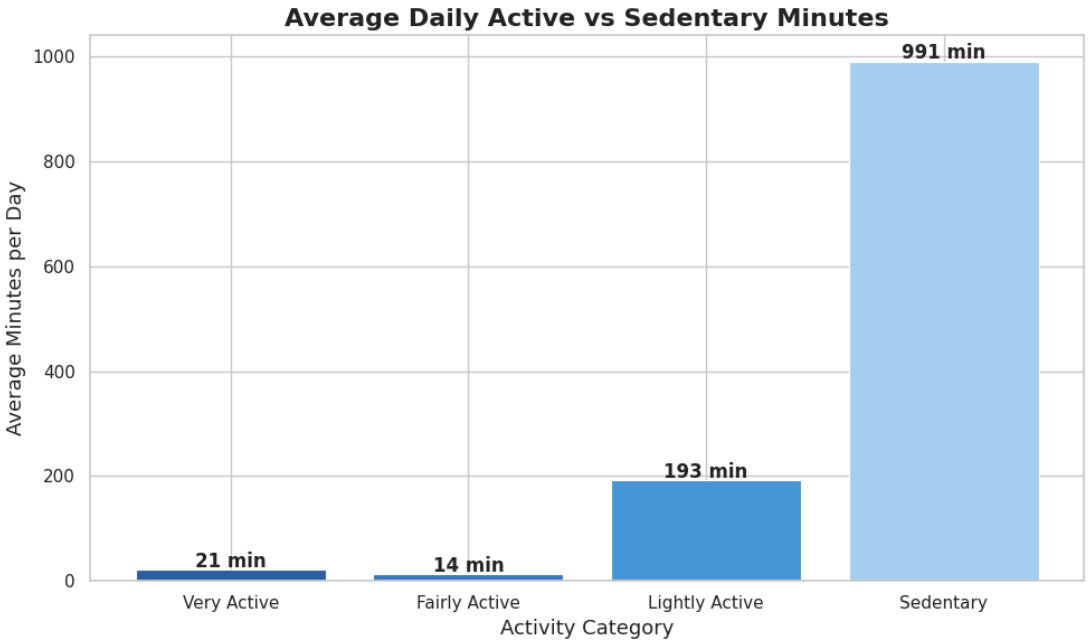
5. Visualizations

All 5 Charts with Explanations

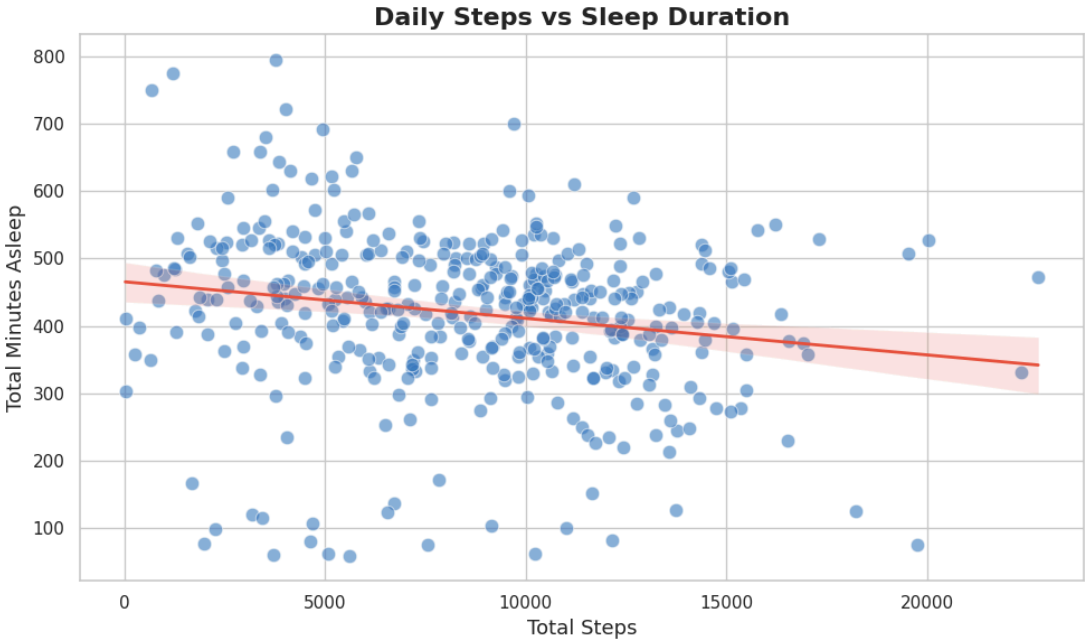
1. **Pie Chart: User Classification by Daily steps.**



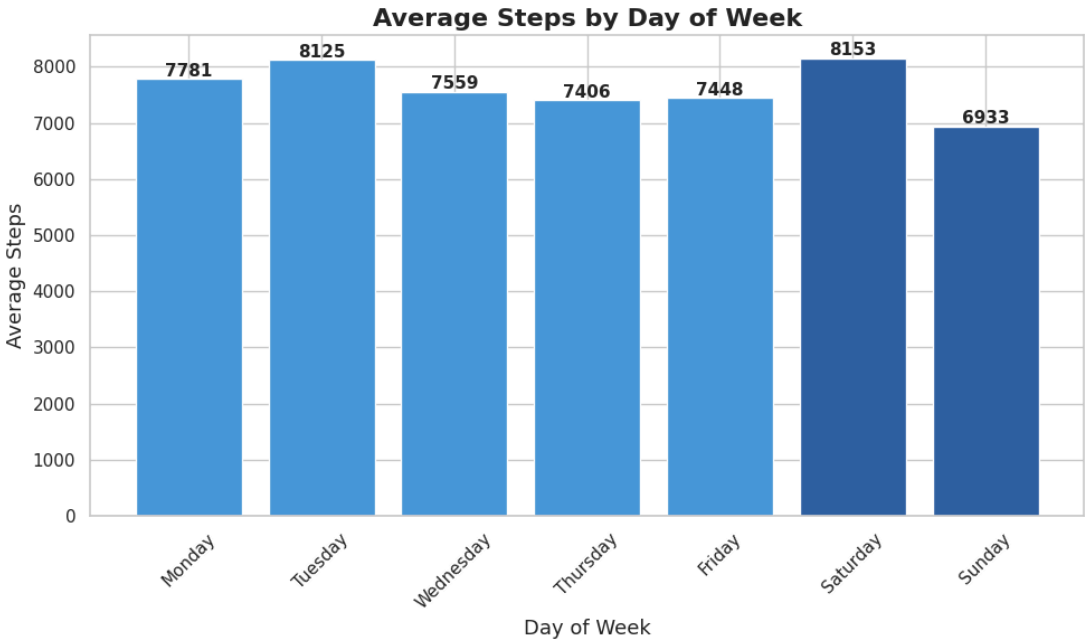
2. **Bar Chart: Average daily steps vs Sedentary Minutes**



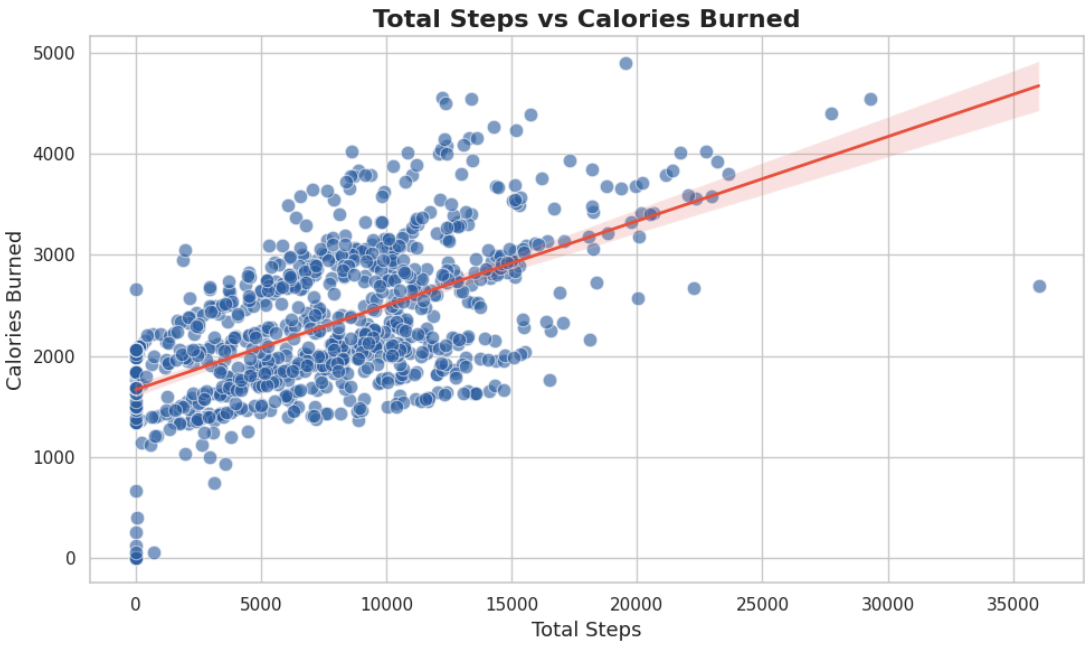
3. **Scatter Plot: Daily Steps vs Sleep Duration**



4. **Bar Chart: Average steps Day of the week**



5. **Scatter Plot : Total steps vs Calories Burned**



6. Top 3 Recommendations (Act Phase)

Based on the analysis, here are the top three actionable recommendations for Bellabeat's marketing and product development:

Recommendation 1 — Market Leaf as a Sedentary Lifestyle Solution

Insight: 51.5% of smart device users fall into Sedentary or Lightly Active categories, with an average of 991 minutes per day spent inactive despite owning a smart device.
Strategy:
Position the Leaf in marketing campaigns as a daily movement companion not just a fitness tracker
Use messaging like "Every step counts — let Leaf remind you to move"
Target working women who sit at desks all day through Google Search and Instagram ads
Highlight the Leaf's gentle vibration reminders feature to encourage movement breaks

Recommendation 2 — Launch a Weekend Wellness Campaign

Insight: Sunday is the least active day with only 6,933 average steps — significantly below weekday averages

Strategy:
Create a Weekend Wellness Challenge campaign through the Bellabeat app
Send push notifications through the app every Sunday morning to motivate Leaf users
Market on Facebook and Instagram on weekends with motivational content
Use messaging like "Don't let Sunday slow you down — your Leaf is ready when you are"

Recommendation 3 — Promote Leaf as a Sleep & Activity Balance Tool

Insight: Highly active users tend to sleep slightly less — showing a tradeoff between activity and rest

Strategy:
Market the Leaf's sleep tracking feature alongside its activity tracking
Position Bellabeat as the brand that understands whole body wellness — not just fitness
Target women through YouTube video ads showing the connection between activity, sleep and overall health
Use messaging like "Move well. Sleep well. Live well — with Leaf"

Marketing Focus: Design content that links the Leaf's activity tracking to its sleep tracking features. The message should be: "The better you move during the day, the better you rest at night." This reinforces the holistic wellness appeal of the Bellabeat Leaf.